

ESC201: Mechanics of Materials

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Homework 8

Instructors: Professor Wootton

Time Spent: 4 Hours

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Note that although all numbers present in this homework assignment are displayed with 2-4 significant digits, no rounding occurs until the final answer.

1. Problem 6.3

Text 4th edition, Ch 6: 3, 15, 31, 46

Problem 6.31 will be graded

6.3 A square box beam is made of two 20 x 80 mm planks and two 20 x 120 mm planks nailed together as shown. Knowing that the spacing between the nails is $s = 50$ mm, and that the allowable shearing force in each nail is 300 N, determine (a) the largest allowable vertical shear in the beam, (b) the corresponding maximum shearing stress in the beam.

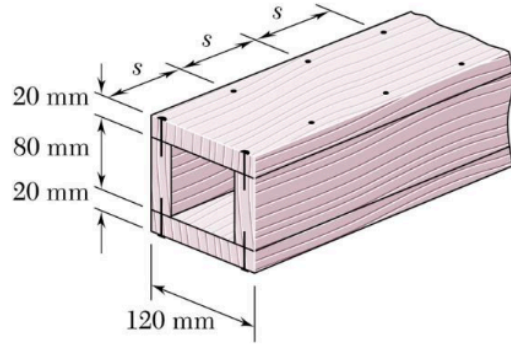


Figure 1: Problem Statement for Problem 6.3

The following given figures are as follows:

$$l_1 = l_2 = 0.020m, \quad w_1 = 0.080m, \quad s = 0.050m, \quad w_2 = 0.120m, \quad \max(V) = 300N$$

$$t = 2l_1 = 0.040m$$

Determine:

- $\max(V)$
- $\max(\tau)$

Approach:

1. Find the moment of inertia I via superposition of the square box beam.
2. Consider the top plank ($l_2 \times w_2$) of the square beam. Assuming the square box beam is of uniform density, find Q using $Q = A_1 \bar{Y}_1$, where $\bar{Y}_1$ is the distance from the neutral axis of the top plank and the neutral axis of the beam, and A_1 is the area of the top plank.
3. Find the maximum vertical shearing force exerted by the nail q_{nail} using $(q = \frac{\Delta F}{\Delta x})$.
4. Rearrange the other q equation $(q_{\text{nail}} = \frac{VQ}{I})$ to find $\max(V)$.
5. Find the first moment of the beam by adding the first moments of half of each of the side planks.
6. Find $\tau_{\max}$ using $(\tau_{\max} = \frac{VQ}{It})$.

Recall that the moment of inertia of a rectangle is $(I = \frac{1}{12}bh^3)$. Therefore, the moment of inertia of this square box beam is:

$$I = \frac{1}{12}w_2(2l_1 + w_1)^3 - \frac{1}{12}(w_2 - 2l_1)w_1^3 \quad (1)$$

Assume the square box beam is of uniform density. Therefore, it's central axis lies at the geometric midpoint of the beam, ($\bar{Y} = 60\text{mm}$). Additionally, the central axis of the top plank is $\bar{Y}' = 10\text{mm}$

Define ($\bar{Y}_1 := \bar{Y} - \bar{Y}'$). Then:

$$Q_1 = A_1 \bar{Y}_1 \quad (2)$$

Since there are two sets of nails on each side of the top plank of the beam, the max q_{nail} corresponds with the allowable shearing force in each nail, i.e.:

$$q = \frac{\Delta F}{\Delta x} = \frac{2V}{s} \quad (3)$$

Lastly:

$$\left(q_{\text{nail}} = \frac{VQ_1}{I} \right) \Rightarrow \left(V = \frac{q_{\text{nail}}I}{Q_1} = \frac{2V \left(\frac{1}{12}w_2(2l_1 + w_1)^3 - \frac{1}{12}(w_2 - 2l_1)w_1^3 \right)}{sA_1 \bar{Y}_1} \right) \quad (4)$$

$$V = 1386.667 \text{ N} \quad (5)$$

Now, to find the first moment of inertia of the beam, use:

$$Q = Q_1 + 2 \left(\frac{w_2}{2} \right) (l_1) \bar{Y}_2 \quad (6)$$

where $\bar{Y}_2$ is the neutral axis of the upper half of the side plank ($\bar{Y}_2 = 20 \text{ mm}$).

Therefore, since the max shearing stress is associated with the maximum shearing force exerted on the nails:

$$\max(\tau) = \frac{VQ}{It} = 3.8 * 10^5 \text{ Pa} \quad (7)$$

2. Problem 6.15

6.15 For the beam and loading shown, (a) sketch profiles of normal and shear stresses on a beam cross-section, and (b) determine the minimum required depth h , knowing that for the grade of timber used, $\sigma_{all} = 1750$ psi in the direction of the wood grain, and $\tau_{all} = 130$ psi for shear stress parallel to the wood grain.

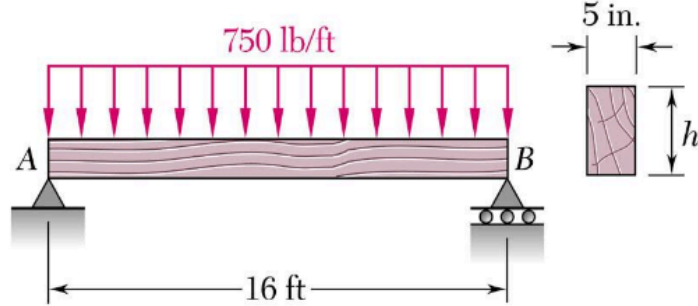


Figure 2: Problem Statement for Problem 6.15

The following given figures are as follows:

$$\sigma_{all} = 1750 \text{ psi}, \quad \tau_{all} = 130 \text{ psi}, \quad l = 16 \text{ ft}, \quad w = 5 \text{ in} = \frac{5}{12} \text{ ft}, \quad p = 750 \text{ lb/ft}$$

Determine: $\min(h)$

Approach:

1. Since the load is equally distributed across the beam, let $(P = pl)$.
2. Note that the maximum horizontal shear force R would occur at either A or B, and use it to calculate the maximum internal bending moment $(M_{max} = \frac{1}{2}(\frac{l}{2})(R))$.
3. Find the necessary depth to accomodate the bending stress of the beam h_b using $(\frac{M_{max}}{\sigma_{all}} = S = \frac{1}{6}bh_b^2)$.
4. Find the necessary depth to accomodate the shear stress of the beam h_s using $(\tau_{max} = \frac{3RQ}{Ib})$.
5. Set $(\min(h) = \max(\{h_b, h_s\}))$.

First, let:

$$P = wl \tag{8}$$

Note that the maximum horizontal shear force R would occur at either A or B, and since p is equally distributed across the beam $(R = \frac{P}{2})$. Therefore (note M_{max} needs to be converted to $\text{lb} \cdot \text{in}$):

$$M_{\max} = \frac{1}{2} \left(\frac{l}{2} \right) (R) \quad (9)$$

Now, from the 3rd step in our approach:

$$\left(\frac{M_{\max}}{\sigma_{\text{all}}} = S = \frac{1}{6} w h_b^2 \right) \Rightarrow \left(h_b = \sqrt{\frac{6 M_{\max}}{w \sigma_{\text{all}}}} = 14.053 \text{in} \right) \quad (10)$$

Now, if we were to treat the upper half of the beam (from its front view) as a rectangle of uniform density, we can calculate its first moment about its neutral axis, as well as its moment of inertia:

$$Q = A \bar{y} = \frac{1}{2} (w) (h_s) \left(\frac{1}{4} h_s \right) = \frac{1}{8} w h_s^2 \quad (11)$$

$$I = \frac{1}{12} w h_s^3 \quad (12)$$

Additionally, from the 4th step in our approach:

$$\left(\tau_{\max} = \frac{3 R Q}{I b} = \frac{3 R}{2 w h_s} \right) \Rightarrow \left(h_s = \frac{3 R}{2 w \tau_{\max}} = 13.846 \text{in} \right) \quad (13)$$

Therefore:

$\min(h) = \max(\{h_b, h_s\}) = 14.053 \text{in} \quad (14)$
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3. Problem 6.31

6.31 The built-up box beam shown at left is made by gluing together two 20 x 250 mm plywood strips and two 50 x 100 mm planks. Knowing that the allowable average shearing stress in the glued joints is 350 kPa, determine the largest permissible vertical shear in the beam.

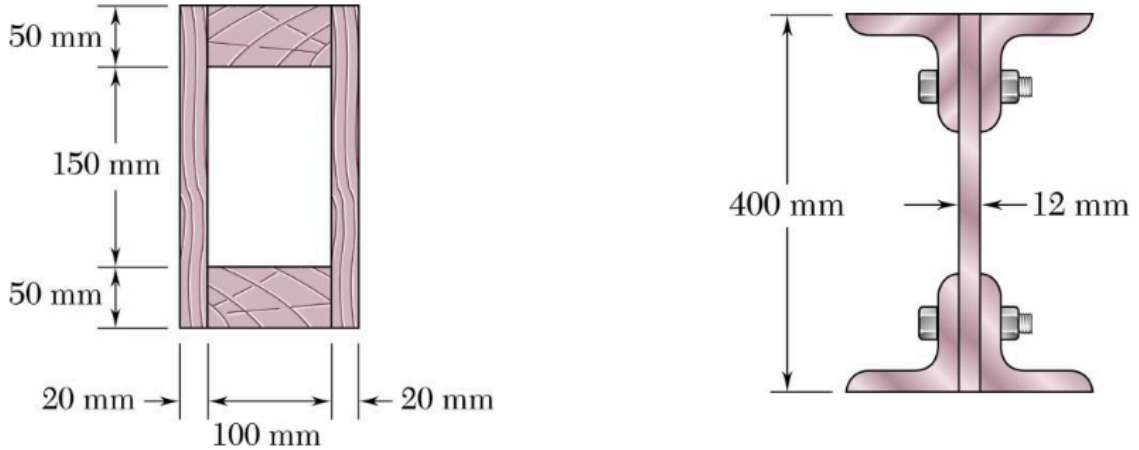


Figure 3: Problem Statement for Problem 6.31

The following given figures are as follows:

$$l_1 = 0.100\text{m}, \quad l_2 = 0.150\text{m}, \quad w_1 = 0.020\text{m}, \quad w_2 = 0.050\text{m}, \quad t = 0.100\text{m}$$

$$\tau_{\max} = 350\text{kPa} = 350000\text{Pa}$$

Determine: $\max(V)$

Approach:

1. Find the moment of inertia of the beam I via superposition.
2. Assuming that the beam is of uniform density, find its first moment Q about its neutral axis using ($Q = A\bar{y}$).
3. Find $\max(V)$ by rearranging ($\tau_{\max} = \frac{V_{\max}Q}{It}$).

Recall that the moment of inertia of a rectangle is ($I = \frac{1}{12}bh^3$). Then:

$$I = \frac{1}{12}((2w_1 + l_1)(2w_2 + l_2)^3 + l_1l_2^3) \quad (15)$$

Assume that the beam is of uniform density. Then:

$$Q = A\bar{y} = l_1w_2\left(\frac{w_2 + l_2}{2}\right) \quad (16)$$

Therefore:

$$\left(\tau_{\max} = \frac{V_{\max} Q}{It} \right) \Rightarrow \left(\max(V) = \frac{\tau_{\max} It}{Q} = 1.079 * 10^4 \text{ N} \right) \quad (17)$$

Discussion: The shear load path is as follows:

1. The top plank experiences a longitudinal compressive force (that differs as you move along the length of the beam).
2. The glue joints transfer that compressive force into the side strips.
3. The side strips carry that transferred force downward to the bottom glue joints.
4. The bottom glue joints transfer that force into the bottom plank.

4. Problem 6.46

6.46 Four 102 x 102 x 9.5 steel angle shapes and a 12 x 400 mm steel plate are bolted together to form a beam with the cross-section shown on the right. The bolts are of 22-mm diameter and are spaced longitudinally every 120 mm. Knowing that the beam is subjected to a vertical shear of 240 kN, determine the average shearing stress in each bolt.

Figure 4: *Problem Statement for Problem 6.46*

The following given figures are as follows:

$$l_1 = w_1 = 0.102\text{m}, \quad d = 9.5\text{m}, \quad V = 240\text{kN}, \quad s = 0.012\text{m}$$

$$l_2 = 0.012\text{m}, \quad w_2 = 0.400\text{m}$$

Determine:

Approach:

1. Considering this angle as two rectangles so we can find the area relevant to the problem.
2. Calculate the centroid of the angles by taking the weighted average of its dimensions relative to its areas.
3. Find the moment of inertia of the angle by using parallel axis theorem and superposition.
4. Find the moment of inertia of the plate using the formula for the moment of inertia of a rectangle.
5. Find the moment of inertia of the object by adding the moments of inertia of the angle and the plate.
6. Find the first moment of the object via $(Q = A\bar{y})$.
7. Find the force applied to the bolt using $(F_{\text{bolt}} = qs)$, where $(q = \frac{VQ}{I})$.

Then, calculate (t_{bolt}) using $(t_{\text{bolt}} = \frac{F_{\text{bolt}}}{A_{\text{bolt}}})$. First, we can consider the angle as two rectangles. The area of the angle is:

$$A = A_1 + A_2 - A_3 = 2l_1d - d^2 \quad (18)$$

The centroid of the angle can be found by taking the weighted average of its dimensions relative to its areas:

$$\bar{y} = \frac{(l_1 d)(\frac{d}{2}) + (l_1 d)(\frac{w_1}{2}) - (d^2)(\frac{d}{2})}{A} \quad (19)$$

Now, we can find the moment of inertia of the angle by using parallel axis theorem and superposition:

$$I_a = \left(\frac{(102)(9.5)^3}{12} + A_1 \left(\frac{d}{2} - \bar{y} \right)^2 \right) + \left(\frac{(102)(9.5)^3}{12} + A_2 \left(\frac{w_1}{2} - \bar{y} \right)^2 \right) - \left(\frac{(9.5)(9.5)^3}{12} + A_3 \left(\frac{d}{2} - \bar{y} \right)^2 \right) \quad (20)$$

Then, find the moment of inertia of the plate using the formula for the moment of inertia of a rectangle:

$$I_{\text{plate}} = (w_2) \frac{(l_2)^3}{12} \quad (21)$$

Now, we can find the moment of inertia of the object by adding the moments of inertia of the angle and the plate:

$$I = I_a + I_{\text{plate}} \quad (22)$$

Next, we can find the first moment of the object via ($Q = A\bar{y}$):

$$Q = A \left(\frac{w_2}{2} - \bar{y} \right) \quad (23)$$

$$F_{\text{bolt}} = qs = \frac{VQs}{I} \quad (24)$$

Finally, we can calculate (τ_{bolt}) using ($\tau_{\text{bolt}} = \frac{F_{\text{bolt}}}{A_{\text{bolt}}}$):

$$\bullet \left(\frac{((9.5)(9.5)^3) / 12 + A_3(d/2 - \overline{y})^2}{\left(\frac{(102)(9.5)^3}{12} + A_1 \left(\frac{d}{2} - \bar{y} \right)^2 + \left(\frac{(102)(9.5)^3}{12} + A_2 \left(\frac{w_1}{2} - \bar{y} \right)^2 - \left(\frac{(9.5)(9.5)^3}{12} + A_3 \left(\frac{d}{2} - \bar{y} \right)^2 \right) \right)} \right) \left(\frac{w_2}{2} - \bar{y} \right) \right)$$

$$\tau_{\text{bolt}} = \frac{F_{\text{bolt}}}{A_{\text{bolt}}} = \frac{VQs}{IA_{\text{bolt}}} = 8.33 * 10^6 \text{ Pa} \quad (25)$$

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5. Appendix Code

The following Jupyter Notebook (attached to the homework submission at “ESC201-HW8.ipynb”) was used for calculations.

```
In [8]: from sympy.solvers import solve
        from sympy import Symbol
        import numpy as np
```

```
In [7]: # Question 6.3

l1 = 0.020
w1 = 0.080
s = 0.050
w2 = 0.120
V = 300
t = 2*l1

I = (w2**4 - w1**4) / 12
A1 = w2 * l1
y1 = 0.050
Q1 = A1 * y1
q = (2*V) / s
V = (q*I) / Q1
Q = Q1 + (l1**2)*w1
tau = (V*Q) / (I*t)

print(f"l1 = {l1:.3f}")
print(f"w1 = {w1:.3f}")
print(f"s = {s:.3f}")
print(f"w2 = {w2:.3f}")
print(f"V = {V:.3f}")
print(f"t = {t:.3f}")
```

```
print(f"I = {I:.6f}")
print(f"A1 = {A1:.3f}")
print(f"y1 = {y1:.3f}")
print(f"Q1 = {Q1:.3f}")
print(f"q = {q:.3f}")
print(f"Q = {Q:.3f}")
print(f"tau = {tau:.3f}")

l1 = 0.020
w1 = 0.080
s = 0.050
w2 = 0.120
V = 1386.667
t = 0.040
I = 0.000014
A1 = 0.002
y1 = 0.050
Q1 = 0.000
q = 12000.000
Q = 0.000
tau = 380000.000
```

In [11]: # Question 6.15

```
sigma = 1750.0
tau = 130.0
l = 16.0
w = 5.0
p = 750.0

P = p*l
R = P / 2
M = (1/4)*l*R*12
```

```
hb = np.sqrt((6*M) / (w * sigma))
hm = (3*R) / (2*w*tau)
```

```
print(f"P = {P:.3f}")
print(f"R = {R:.3f}")
print(f"M = {M:.3f}")
print(f"hb = {hb:.3f}")
print(f"hm = {hm:.3f}")
```

```
P = 12000.000
```

```
R = 6000.000
```

```
M = 288000.000
```

```
hb = 14.053
```

```
hm = 13.846
```

In [15]: # Question 6.36

```
l1 = 0.100
```

```
l2 = 0.150
```

```
w1 = 0.020
```

```
w2 = 0.050
```

```
t = 0.100
```

```
tau = 350000.0
```

```
I = ((2*w1 + l1)*((2*w2 + l2)**3) - l1*(l2**3)) / 12
```

```
Q = (l1*w2*(w2+l2)) / 2
```

```
V = (I*t*tau) / Q
```

```
print(f"I = {I:.6f}")
```

```
print(f"Q = {Q:.6f}")
```

```
print(f"V = {V:.3f}")
```

$$I = 0.000154$$

$$Q = 0.000500$$

$$V = 10791.667$$